FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 3

PHYSICS / APPLIED PHYSICS

ELECTRODYNAMICS II

Propagation of Electromagnetic Waves in Linear Media, Transient Circuits and AC Generator

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	Write the expressions for the velocity v and index of refraction n of an electromagnetic wave in a linear medium of permittivity ε and permeability μ. How are the field amplitudes B_0 and E_0 related in that medium?	**3**
2	State Brewster's law for the angle θ_B at which the reflected wave is extinguished, for $\mu_1 \approx \mu_2$. Find θ_B for light passing from air ($n_1 = 1$) into glass ($n_2 = 1.5$), and state the polarization of the reflected beam at θ_B.	**3**
3	Give an expression for the instantaneous current during the growth of current in a series L-R circuit. What are the terms involved? State whether the time taken by the current to reach its maximum value increases or decreases if L/R increases.	**3**
4	Distinguish between initiation and transition transient currents.	**3**
5	Write the expression for the instantaneous current in an L-R circuit after the cell is removed and the circuit is closed on itself, if the current at the instant of removal is I_0. Define the time constant of decay.	**3**
6	Write down the equation for the decay of charge on a capacitor C discharging through a resistor R. How does a smaller time constant CR affect the rate of discharge of the capacitor?	**3**
7	A capacitor C charged to a potential V is connected across an inductor L of negligible resistance at $t = 0$. Write the differential equation for the circuit and give the expression for its natural frequency ν in terms of L and C.	**3**
8	A square loop of side length a is mounted on a vertical shaft and rotated at constant angular velocity ω in a uniform magnetic field $\mathbf{B}$ perpendicular to the shaft. Find the induced e.m.f. $\mathcal{E}(t)$ for this alternating current generator.	**3**
9	Write the expression for the instantaneous e.m.f. and peak value E_0 of the e.m.f. induced in a coil of N turns and area A rotating with angular velocity ω in a uniform magnetic field B. State the orientations of the coil for which the induced e.m.f. is zero and maximum.	**3**
10	Write the mean value of an alternating current over half a cycle and the r.m.s. value in terms of its peak value I_0. What is the ratio of the r.m.s. value of current to the mean value of current called, and what is its numerical value?	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	Give the fundamental laws of geometrical optics considering the reflection and transmission of electromagnetic waves at a boundary separating two linear media.	**6**
12	A monochromatic plane wave of frequency $5 \times 10^{14}\ \text{Hz}$ passes from air ($n_1 = 1.00$) into glass of refractive index $n_2 = 1.50$ (take $\mu = \mu_0$). Find its speed in glass, its wavelength in vacuum, and its wavelength in glass. Also find the fraction of the incident intensity that is reflected and the fraction transmitted at normal incidence.	**6**
13	A coil of inductance $0.5\ \text{H}$ and resistance $10\ \Omega$ is connected to a $12\ \text{V}$ cell. Find the time constant, the final steady current, the instantaneous current after $0.1\ \text{s}$, and the initial rate of growth of current at $t = 0$. Take $e^{-2} = 0.1353$.	**6**
14	An uncharged capacitor C is connected in series with a resistor R and a battery of voltage V_0 at time $t = 0$. Obtain expressions for the growth of charge $Q(t)$ and current $I(t)$. Find the total energy supplied by the battery, the heat delivered to the resistor, and the fraction of the work done by the battery that shows up as energy in the capacitor.	**6**
15	A capacitor of $3\ \mu\text{F}$ is discharged through a resistance. The time taken for half the charge on the capacitor to leak is found to be 5 **seconds**. Compute the value of the resistance. If a capacitor of $0.2\ \mu\text{F}$ is first charged and discharged through a resistance of $10\ \text{M}\Omega$, find the time the potential takes to fall to $\frac{1}{4}$ of its original value.	**6**
16	A capacitor C charged to potential V_0 is discharged through an inductance L and resistance R in series. Set up the differential equation for the discharge and plot the graphs showing the relation	**6**

	between charge and time in (a) critically damped, and (b) oscillatory discharge of the capacitor. Give the conditions in each case.	
17	Derive the expression for the e.m.f. induced in a coil rotating with uniform angular velocity in a uniform magnetic field. Hence obtain the equations for the r.m.s. value of e.m.f. and current in an AC circuit.	**6**
18	A rectangular coil of 100 turns and area $0.03\ \mathrm{m}^2$ rotates at $1200\ \mathrm{r.p.m.}$ about an axis perpendicular to a uniform magnetic field of $0.4\ \mathrm{T}$. Find (a) the peak value of the induced e.m.f., (b) the r.m.s. and half-cycle mean values of the induced e.m.f., and (c) the instantaneous e.m.f. when the plane of the coil makes an angle of $30°$ with the magnetic field.	**6**
Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	In a linear homogeneous medium with no free charge or current, derive the wave equations for $\vec{E}$ and $\vec{B}$ from Maxwell's equations and show that $v = 1/\sqrt{\varepsilon\mu} = c/n$. Hence find expressions for the reflection and transmission coefficients when a plane polarized monochromatic wave of angular frequency ω passes normally through the boundary between two linear dielectrics.	**10**
20	Discuss the growth of current in a circuit containing an inductance L and a resistance R connected in series with a cell of steady e.m.f. Explain the time constant of the circuit and show graphically how the inductance affects the rate of growth of current.	**10**